(ADAPTER)User Story 8 – Logging, Monitoring, and Traceability for P2P Flows

Story ID:US-ADAPTER-P2P-008
Actor: Adapter Integration (primary), IPL and UCRM operations teams (consumers of logs/metrics)
Goal: Provide end-to-end traceability and monitoring for P2P message flows in both directions at the integration layer.
Business Value: Facilitates troubleshooting, compliance, and operational monitoring of P2P communications between IPL and UCRM, without exposing internal UCRM processing details.
Referenced Diagram:UCRI 2.0 – P2P FlowDiagram (overall flow; traceability across all integration steps).
Jira ticket: IPL-26522: (ADAPTER) Logging, Monitoring, and Traceability for P2P Flows OPEN

User Story:

> As an operations stakeholder, I want the Adapter to log and expose key events and statuses for P2P message flows so that I can monitor, trace, and troubleshoot IPL ↔ UCRM communications.

Key Scenarios & Variations:

- Logging of successful flows (IPL→UCRM and UCRM→IPL).
- Logging of validation, authentication, and forwarding failures.
- Correlation of logs across IPL, Adapter, and UCRM using identifiers.

Acceptance Criteria:

- Given a message processed through the Adapter (in either direction)
  When the flow completes (success or failure)
  Then the Adapter must log key events: reception, validation result, authentication attempts, forwarding attempts, UCRM/IPL response, and final status.
- Given a correlation ID associated with a message
  When operations search logs
  Then they must be able to reconstruct the end-to-end flow across IPL, Adapter, and UCRM using that ID (to the extent UCRM logs are available).

- Given repeated failures (e.g., authentication failures, delivery failures)
  When thresholds are exceeded
  Then the Adapter must expose metrics or alerts so that operations can intervene.

